

D901

Secondary Mathematics Curriculum

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A standard says students solve systems of linear equations algebraically and graphically and explain why the solutions match. A teacher finds a free worksheet with 2 million downloads that has 30 systems solved by substitution only. How well does it align?

- A. Fully, because the topic is systems of equations.
- B. Partly: it covers algebraic solving but omits graphing and explanation.
- C. Fully, because so many teachers have downloaded it.
- D. Not at all, because worksheets never align to standards.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A teacher preparing a number-sort for a class that meets in room 214 lists four numbers. The sort will be printed on cardstock. Which number is irrational?

- A. 0.121212...
- B. $\frac{22}{7}$
- C. $\sqrt{50}$
- D. $\sqrt{49}$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

Which statement is always true?

- A. The product of two irrational numbers is irrational.
- B. The sum of a rational number and an irrational number is irrational.
- C. The sum of two irrational numbers is irrational.
- D. The product of a rational number and an irrational number is irrational.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

On a review day before a three-day weekend, a teacher asks for the multiplicative inverse of $-3/4$. The review is projected from a document camera. What is it?

- A. $-4/3$
- B. $4/3$
- C. 1
- D. $3/4$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

Which property does $7 + (-7) = 0$ illustrate?

- A. Multiplicative inverse.
- B. Additive identity.
- C. Additive inverse.
- D. Commutative property of addition.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A ball rolls down a ramp built by the robotics club; its position is $s(t) = t^2 + 2t$ meters after t seconds. The ramp is 4 meters long. What is its instantaneous velocity at $t = 3$?

- A. 8 m/s
- B. 5 m/s
- C. 15 m
- D. 6 m/s

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

A function has $f(2) = 5$ and $f'(2) = 3$. What is the equation of the tangent line at $x = 2$?

- A. $y = 5x - 7$
- B. $y = 3x + 11$
- C. $y = 3x + 5$
- D. $y = 3x - 1$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A geometry teacher who keeps a class set of protractors writes: "If a quadrilateral is a square, then it is a rectangle." The statement is true. What is its converse, and is it true?

- A. The converse is true, because the converse of a true statement is always true.
- B. "If a quadrilateral is not a rectangle, then it is not a square." True.
- C. "If a quadrilateral is not a square, then it is not a rectangle." False.
- D. "If a quadrilateral is a rectangle, then it is a square." False.

Target word _____	The one fact that decides _____	Answer ___	[] sure [] unsure [] guess
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9.

On a 12-mile commute, the chance of a red light at the first intersection is 0.4 and at the second is 0.5, independently. The commute takes about 25 minutes. What is the probability of red lights at both?

- A. 0.20
- B. 0.90
- C. 0.45
- D. 0.70

Target word _____	The one fact that decides _____	Answer ___	[] sure [] unsure [] guess
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10. [TRANSFER — reversed stem / different variable]

Events A and B are independent. $P(A \text{ and } B) = 0.12$ and $P(A) = 0.3$. What is $P(B)$?

- A. 0.036
- B. 0.4
- C. 0.18
- D. 0.88

Target word _____	The one fact that decides _____	Answer ___	[] sure [] unsure [] guess
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11.

A math department with a \$40,000 budget reviews three algebra programs. One has glossy design and a free trial; another has independent reviews showing strong standards alignment and coherence. What should the committee weigh first?

- A. Whichever has the most practice problems.
- B. Which program looks most appealing and offers a free trial.
- C. Which program is cheapest.
- D. Alignment to standards, coherence across grades, and evidence of effectiveness.

Target word _____	The one fact that decides _____	Answer ___	[] sure [] unsure [] guess
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12.

A teacher wants students to discover the triangle angle-sum relationship. The school recently upgraded its Wi-Fi. Which use of technology best serves the goal?

- A. Have students add given angle measures with a calculator.
- B. Show a slideshow that states the theorem.
- C. Students drag triangle vertices in a dynamic-geometry tool, watch the angle sum, conjecture, then prove it.
- D. Any technology will improve the lesson.

Target word _____	The one fact that decides _____	Answer ____	[] sure [] unsure [] guess
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STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A standard says students solve systems of linear equations algebraically and graphically and explain why the solutions match. A teacher finds a free worksheet ~~with 2 million downloads~~ that has 30 systems solved by substitution only. How well does it align?

Target word: degree of alignment **Decides:** The worksheet covers only one method and no explanation.

A	Dropped element	Ignores the graphing and explanation parts of the standard.
B	Correct	Topic matches, but the standard's full expectation does not.
C	Camouflage	Popularity is not alignment.
D	Absolute language	Format does not decide alignment; content and rigor do.

2. Answer C Concept 2

A teacher preparing a number-sort ~~for a class that meets in room 214~~ lists four numbers. ~~The sort will be printed on cardstock.~~ Which number is irrational?

Target word: classify a number **Decides:** $\sqrt{50} = 5\sqrt{2}$, and $\sqrt{2}$ is not a ratio of integers.

A	Adjacent category	A repeating decimal equals $12/99$, a rational number.
B	Confusion set	Only an approximation of π ; it is a ratio of integers.
C	Correct	50 is not a perfect square; $\sqrt{50} = 5\sqrt{2}$ is irrational.
D	Dropped element	$\sqrt{49} = 7$; a radical sign alone does not make a number irrational.

3. Answer B Concept 2 • transfer

Which statement is always true?

Target word: closure of rational and irrational numbers **Decides:** Rational + irrational cannot be rational.

A	Absolute language	$\sqrt{2} \times \sqrt{2} = 2$.
B	Correct	If it were rational, subtracting the rational part would make the irrational number rational.
C	Confusion set	$\sqrt{2} + (-\sqrt{2}) = 0$.
D	Dropped element	Fails when the rational number is 0.

4. Answer A Concept 3

On a review day ~~before a three-day weekend~~, a teacher asks for the multiplicative inverse of $-3/4$. ~~The review is projected from a document camera.~~ What is it?

Target word: multiplicative inverse **Decides:** $(-3/4) \times (-4/3) = 1$.

A	Correct	Its product with $-3/4$ is 1.
B	Dropped element	Drops the sign; the product would be -1 .
C	Adjacent category	1 is the multiplicative identity, not the inverse.
D	Confusion set	That is the additive inverse, which sums to 0.

5. Answer C Concept 3 · transfer

Which property does $7 + (-7) = 0$ illustrate?

Target word: name the property **Decides:** A number plus its opposite gives the additive identity.

A	Inverted operation	Multiplicative inverses multiply to 1.
B	Confusion set	The identity property is $a + 0 = a$.
C	Correct	-7 is the additive inverse of 7.
D	Adjacent category	Commutativity is $a + b = b + a$.

6. Answer A Concept 4

A ball rolls down a ramp ~~built by the robotics club~~; its position is $s(t) = t^2 + 2t$ meters after t seconds. ~~The ramp is 4 meters long.~~ What is its instantaneous velocity at $t = 3$?

Target word: derivative at a point **Decides:** $s'(t) = 2t + 2$.

A	Correct	$s'(3) = 2(3) + 2 = 8$.
B	Adjacent category	$s(3)/3 = 15/3$ is the average velocity over $[0, 3]$.
C	Confusion set	$s(3)$ is the position, not the velocity.
D	Dropped element	Differentiates t^2 but drops the 2 from the $2t$ term.

7. Answer D Concept 4 · transfer

A function has $f(2) = 5$ and $f'(2) = 3$. What is the equation of the tangent line at $x = 2$?

Target word: tangent line from f and f' **Decides:** $y - 5 = 3(x - 2)$.

A	Confusion set	Swaps the value and the slope.
B	Inverted operation	Uses $x + 2$ instead of $x - 2$.
C	Dropped element	Uses $f(2)$ as the y-intercept, skipping the point-slope step.
D	Correct	Slope 3 through $(2, 5)$: $y = 3x - 6 + 5$.

8. Answer D Concept 5

A geometry teacher ~~who keeps a class set of protractors~~ writes: "If a quadrilateral is a square, then it is a rectangle." The statement is true. What is its converse, and is it true?

Target word: form and truth of the converse **Decides:** The converse swaps hypothesis and conclusion.

A	Absolute language	Converses can be false.
B	Adjacent category	That is the contrapositive.
C	Confusion set	That is the inverse, not the converse.
D	Correct	A 2-by-5 rectangle is not a square.

9. Answer A Concept 6

~~On a 12-mile commute,~~ the chance of a red light at the first intersection is 0.4 and at the second is 0.5, independently. ~~The commute takes about 25 minutes.~~ What is the probability of red lights at both?

Target word: $P(A \text{ and } B)$ **Decides:** Independence lets us multiply.

A	Correct	$0.4 \times 0.5 = 0.20$.
B	Inverted operation	Adds instead of multiplying.
C	Adjacent category	Averages the probabilities.
D	Confusion set	$0.4 + 0.5 - 0.2$ is $P(A \text{ or } B)$.

10. Answer B Concept 6 · transfer

Events A and B are independent. $P(A \text{ and } B) = 0.12$ and $P(A) = 0.3$. What is $P(B)$?

Target word: solve for $P(B)$ **Decides:** $0.12 = 0.3 \times P(B)$.

A	Inverted operation	Multiplies instead of dividing.
B	Correct	$0.12 \div 0.3 = 0.4$.
C	Adjacent category	Subtracts, as if the events were additive.
D	Wrong stem number	$1 - 0.12$ works with the joint probability instead of solving for $P(B)$.

11. Answer D Concept 7

A math department ~~with a \$40,000 budget~~ reviews three algebra programs. One has glossy design and a free trial; another has independent reviews showing strong standards alignment and coherence. What should the committee weigh first?

Target word: first criterion **Decides:** Alignment and coherence come before usability or cost.

A	Confusion set	Volume is not quality or alignment.
B	Camouflage	Design and trials do not show alignment.
C	Adjacent category	Cost matters only among aligned options.
D	Correct	A resource that misses the standards fails no matter its other features.

12. Answer C Concept 8

A teacher wants students to discover the triangle angle-sum relationship. ~~The school recently upgraded its Wi-Fi.~~ Which use of technology best serves the goal?

Target word: purposeful technology **Decides:** Students must test many cases to form a conjecture.

A	Dropped element	Leaves out the exploring that leads to a conjecture.
B	Adjacent category	Technology used only to present; students discover nothing.
C	Correct	The tool lets students test many triangles quickly.
D	Absolute language	Technology helps only when it serves the goal.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	7	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	9	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	4	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	2	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Alignment means content and rigor	A task or resource aligns to a standard only if it matches both the content and the full expectation of the standard (methods, representations, and reasoning).	Calling a resource aligned because the topic matches or because it is popular.	A standard's wording set next to a resource's description.
2	Rational vs. irrational numbers	Rational numbers can be written as a ratio of integers (including repeating decimals and perfect-square roots); irrational numbers cannot.	Calling any repeating decimal or any radical irrational, or treating $22/7$ as π .	A list of numbers to classify.
3	Identity and inverse elements	The additive identity is 0 and the multiplicative identity is 1; an inverse combines with a number to give the identity ($a + (-a) = 0$; $a \cdot (1/a) = 1$).	Mixing up additive and multiplicative inverses, or dropping the sign of a reciprocal.	"Inverse" or "identity" in a property question.
4	The derivative as instantaneous rate of change	$f'(a)$ is the instantaneous rate of change and the slope of the tangent line at $x = a$; it is not the average rate over an interval.	Reporting the position or the average rate instead of the derivative.	"Instantaneous," "at the moment," or "tangent line."
5	Conditional, converse, inverse, contrapositive	Only the contrapositive is logically equivalent to the original statement; the converse swaps hypothesis and conclusion and may be false.	Assuming the converse of a true statement is true.	"If..., then..." and a request for the converse.
6	Independent events multiply	For independent events, $P(A \text{ and } B) = P(A) \times P(B)$.	Adding probabilities for "and."	Two unrelated events that must both happen.
7	Evaluating curricular resources	Judge a resource first on standards alignment, coherence, and evidence of effectiveness; usability and cost matter only among aligned options.	Choosing on appearance, price, or popularity.	A committee comparing programs.
8	Technology that serves the goal	Choose technology when it lets students do mathematics they otherwise could not, such as testing many cases in a dynamic tool to form a conjecture.	Using technology only to display notes, or assuming technology always helps.	A learning goal and several ways to use a tool.

Course scope (public description): This Secondary Mathematics course is designed to prepare future secondary math educators by focusing on the application of fundamental math concepts, the development of effective teaching strategies, and the alignment of instruction with state content standards. This course teaches key areas including application of number systems, algebraic structures, calculus, geometric reasoning, and probability and statistics in the secondary setting. Additionally, learners will learn to design and evaluate mathematical problems, integrate technology, and differentiate instruction to meet all students' needs. Learners will also explore how to research and evaluate curricular resources to ensure they align with educational goals and

standards. By the end of the course, participants will be equipped with the skills and knowledge necessary to deliver high- quality math instruction in secondary education.